

Logistic Regression: A Tutorial with Worked Example and Visualization

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1 Introduction

Logistic Regression is a supervised learning algorithm used for **binary classification**. Although it contains the word “regression,” it is actually a classification algorithm that predicts the probability that an instance belongs to a particular class (e.g., Yes/No, 0/1).

2 Mathematical Foundation

The logistic regression model predicts the probability that the dependent variable Y equals 1 (positive class), given the independent variable X :

$$P(Y = 1|X) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X)}}$$

This is called the **logistic function** or **sigmoid function**. It outputs values between 0 and 1, ideal for probabilities.

2.1 The Sigmoid Function

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

where $z = \beta_0 + \beta_1 X$.

Interpretation:

- If $z \rightarrow +\infty$, then $\sigma(z) \rightarrow 1$
- If $z \rightarrow -\infty$, then $\sigma(z) \rightarrow 0$

Hence, logistic regression maps any real-valued number into a probability range (0, 1).

3 The Logit Function

The **logit** is the natural logarithm of the odds that the dependent variable is 1:

$$\text{logit}(P) = \ln\left(\frac{P}{1 - P}\right) = \beta_0 + \beta_1 X$$

This expresses the relationship between input variables and the log-odds of the outcome.

4 Derivation: From Logit to Logistic Function

We can convert the logit form into a probability expression. Starting with the logit equation:

$$\ln\left(\frac{P}{1 - P}\right) = \beta_0 + \beta_1 X$$

Exponentiating both sides removes the natural log:

$$\frac{P}{1-P} = e^{\beta_0 + \beta_1 X}$$

Multiply both sides by $(1 - P)$:

$$P = (1 - P)e^{\beta_0 + \beta_1 X}$$

Expand and rearrange terms:

$$P = e^{\beta_0 + \beta_1 X} - Pe^{\beta_0 + \beta_1 X}$$

Collect P terms on one side:

$$P(1 + e^{\beta_0 + \beta_1 X}) = e^{\beta_0 + \beta_1 X}$$

Solving for P gives:

$$P = \frac{e^{\beta_0 + \beta_1 X}}{1 + e^{\beta_0 + \beta_1 X}}$$

Alternatively, this can be expressed as the **logistic (sigmoid) function**:

$$P = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X)}}$$

This ensures that the predicted probability P always lies between 0 and 1.

5 Worked Example: Understanding Odds, Log-Odds, and Probability

In logistic regression, the output of the model is expressed as the **log-odds** (also called the **logit**), which represents the natural logarithm of the odds of the event occurring.

$$\text{logit}(p) = \ln\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 X$$

The probability p of the event occurring can be obtained by transforming this equation as:

$$p = \frac{e^{(\beta_0 + \beta_1 X)}}{1 + e^{(\beta_0 + \beta_1 X)}}$$

5.1 Scenario

Suppose we are predicting whether a student receives a job offer ($Y = 1$) based on their CGPA (X). After fitting the logistic regression model, we obtain the following parameters:

$$\beta_0 = -9, \quad \beta_1 = 1.2$$

5.2 Step 1: Compute Log-Odds for a Student with CGPA = 8.5

$$\text{logit}(p) = -9 + 1.2 \times 8.5 = -9 + 10.2 = 1.2$$

So the **log-odds** of getting a job for a student with CGPA = 8.5 is **1.2**.

5.3 Step 2: Convert Log-Odds to Odds

We can find the odds using:

$$\text{odds} = e^{\text{logit}(p)} = e^{1.2} \approx 3.32$$

Interpretation: The odds of getting a job are 3.32 to 1, meaning the student is about 3.3 times more likely to get a job than not.

5.4 Step 3: Convert Odds to Probability

We can convert odds to probability as:

$$p = \frac{\text{odds}}{1 + \text{odds}} = \frac{3.32}{1 + 3.32} = \frac{3.32}{4.32} \approx 0.77$$

Result: The model predicts that this student has a 77% probability of getting a job.

5.5 Step 4: Compute for a Lower CGPA

Now, let's take another student with CGPA = 6.5.

$$\text{logit}(p) = -9 + 1.2 \times 6.5 = -9 + 7.8 = -1.2$$

$$\text{odds} = e^{-1.2} \approx 0.30$$

$$p = \frac{0.30}{1 + 0.30} = 0.23$$

Result: The model predicts that this student has only a 23% probability of getting a job.

5.6 Step 5: Comparative Summary

CGPA	Logit (Log-Odds)	Odds (e^{logit})	Probability (p)
6.5	-1.2	0.30	0.23 (23%)
8.5	1.2	3.32	0.77 (77%)

5.7 Step 6: Interpretation and Insights

- When the **logit (log-odds)** is **positive**, the predicted probability $p > 0.5$, meaning the model predicts a **positive outcome (Job = Yes)**.
- When the **logit** is **negative**, the predicted probability $p < 0.5$, meaning the model predicts a **negative outcome (Job = No)**.

- The **decision boundary** occurs where $\text{logit} = 0$ (i.e., $p = 0.5$).

Interpretation Summary:

- The higher the CGPA, the higher the log-odds and therefore the probability of getting a job.
- The logit function linearly increases with CGPA, while the probability curve (sigmoid) increases non-linearly.

5.8 Visualization Suggestion

You can plot the sigmoid curve to visualize how probability changes with CGPA.

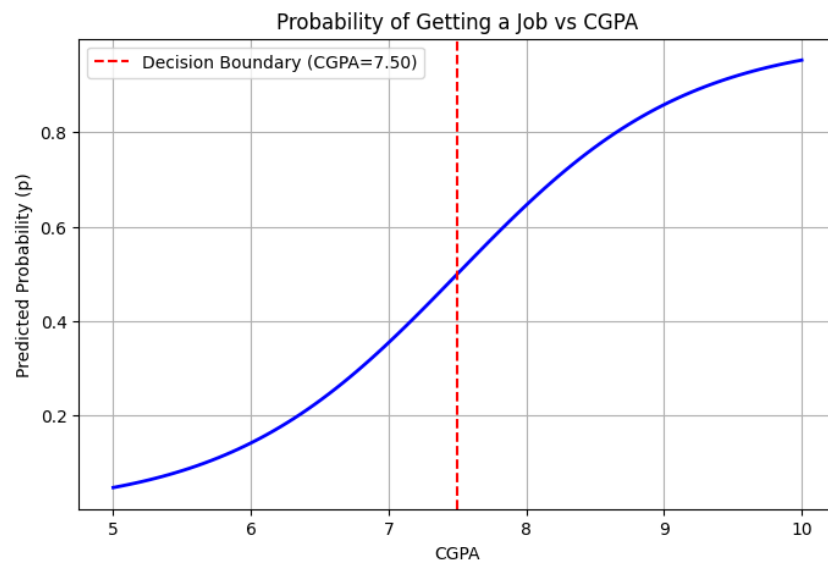


Figure 1: Sigmoid Curve showing relationship between CGPA and probability of getting a job.

6 Worked Example: Predicting Job Offer Based on CGPA

6.1 Dataset

Suppose we want to predict whether a student receives a job offer ($Y = 1$) based on their CGPA (X):

Student	CGPA (X)	Job Offer (Y)
1	8.0	0
2	8.5	0
3	9.0	1
4	9.5	1
5	10.0	1

6.2 Step 1: Model Definition

We define:

$$P(Y = 1|X) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X)}}$$

6.3 Step 2: Parameter Estimation

Parameters β_0 and β_1 are estimated via **Maximum Likelihood Estimation (MLE)**. Assume the model found:

$$\beta_0 = -15, \quad \beta_1 = 1.6$$

6.4 Step 3: Compute Predicted Probabilities

Compute $z = \beta_0 + \beta_1 X$ and the corresponding probabilities.

Student	CGPA (X)	$z = \beta_0 + \beta_1 X$	$P(Y = 1 X)$	Predicted Class
1	8.0	$-15 + 1.6(8.0) = -2.2$	$\frac{1}{1+e^{2.2}} = 0.10$	0 (No)
2	8.5	$-15 + 1.6(8.5) = -1.4$	$\frac{1}{1+e^{1.4}} = 0.20$	0 (No)
3	9.0	$-15 + 1.6(9.0) = -0.6$	$\frac{1}{1+e^{0.6}} = 0.35$	0 (No)
4	9.5	$-15 + 1.6(9.5) = 0.2$	$\frac{1}{1+e^{-0.2}} = 0.55$	1 (Yes)
5	10.0	$-15 + 1.6(10.0) = 1.0$	$\frac{1}{1+e^{-1.0}} = 0.73$	1 (Yes)

6.5 Step 4: Decision Rule

We classify using a threshold of 0.5:

If $P(Y = 1|X) \geq 0.5$, predict $Y = 1$ (Job = Yes)

If $P(Y = 1|X) < 0.5$, predict $Y = 0$ (Job = No)

Thus:

- Students 4 and 5 are predicted to get jobs.
- Students 1, 2, and 3 are predicted to not get jobs.

6.6 Step 5: Decision Boundary

The boundary occurs where $P(Y = 1|X) = 0.5$, which means:

$$\beta_0 + \beta_1 X = 0 \Rightarrow X = -\frac{\beta_0}{\beta_1} = \frac{15}{1.6} = 9.375$$

Therefore:

- If $\text{CGPA} \geq 9.375 \rightarrow \text{Predict Job} = \text{Yes}$
- If $\text{CGPA} < 9.375 \rightarrow \text{Predict Job} = \text{No}$

7 Visualization of the Sigmoid Function

The Sigmoid Curve: The logistic regression model's output follows an S-shaped sigmoid curve. It maps any real-valued input (z) to a probability between 0 and 1.

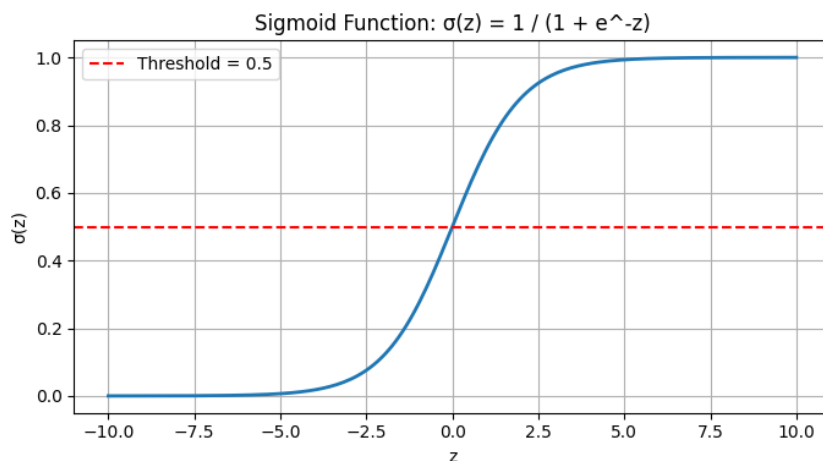


Figure 2: Sigmoid function $\sigma(z) = \frac{1}{1+e^{-z}}$

8 Model Evaluation Metrics

Evaluating a classification model requires understanding several key metrics derived from the **Confusion Matrix**:

	Predicted Positive	Predicted Negative
Actual Positive	True Positive (TP)	False Negative (FN)
Actual Negative	False Positive (FP)	True Negative (TN)

8.1 Accuracy

Accuracy measures the overall correctness of the model:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

Interpretation: Fraction of total predictions that were correct.

8.2 Precision

Precision tells us how many of the predicted positives were actually positive:

$$\text{Precision} = \frac{TP}{TP + FP}$$

Interpretation: High precision means fewer false positives.

8.3 Recall (Sensitivity or True Positive Rate)

Recall measures how many actual positives were correctly identified:

$$\text{Recall} = \frac{TP}{TP + FN}$$

Interpretation: High recall means fewer false negatives.

8.4 F1-Score

The F1-score combines precision and recall using their harmonic mean:

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

Interpretation: Balances the trade-off between precision and recall.

8.5 Example Calculation

Assume we have:

$$TP = 45, \quad TN = 40, \quad FP = 10, \quad FN = 5$$

Then:

$$\text{Accuracy} = \frac{45 + 40}{100} = 0.85$$

$$\text{Precision} = \frac{45}{45 + 10} = 0.818$$

$$\text{Recall} = \frac{45}{45 + 5} = 0.9$$

$$F1 = 2 \times \frac{0.818 \times 0.9}{0.818 + 0.9} = 0.857$$

9 ROC Curve and AUC

9.1 Definition

The **Receiver Operating Characteristic (ROC)** curve visualizes the performance of a binary classifier at all classification thresholds.

It plots:

$$\text{True Positive Rate (TPR)} = \frac{TP}{TP + FN}$$

against

$$\text{False Positive Rate (FPR)} = \frac{FP}{FP + TN}$$

Each threshold (from 0 to 1) gives a point on the ROC curve.

9.2 AUC (Area Under Curve)

The **AUC** represents the probability that the classifier ranks a randomly chosen positive instance higher than a randomly chosen negative one.

$$AUC = \int_0^1 TPR(FPR) d(FPR)$$

Interpretation:

- $AUC = 1.0 \rightarrow$ Perfect classifier
- $AUC = 0.5 \rightarrow$ Random guessing
- $AUC < 0.5 \rightarrow$ Worse than random

9.3 Example Intuition

If we vary the threshold:

- A low threshold $\rightarrow$ More positives predicted (high recall, low precision)
- A high threshold $\rightarrow$ Fewer positives predicted (high precision, low recall)

The ROC curve shows this trade-off across all thresholds. The closer the curve hugs the top-left corner, the better the model.

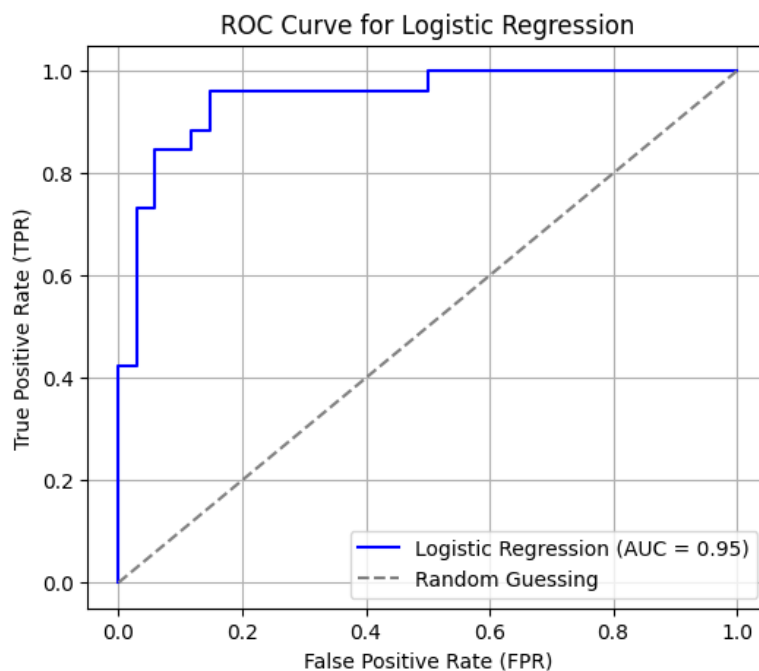


Figure 3: Example ROC Curve showing TPR vs FPR

AUC Interpretation Example:

$AUC = 0.92 \Rightarrow$ The model has a 95% chance of ranking a positive instance higher than a negative one

10 Advantages and Disadvantages

10.1 Advantages

- Easy to implement and interpret.
- Produces probabilities, not just classifications.
- Effective for linearly separable data.

10.2 Disadvantages

- Assumes a linear relationship between independent variables and log-odds.
- Not suitable for non-linear data without transformation.
- Can overfit if data has many correlated features.

11 Conclusion

Logistic Regression is one of the most fundamental and interpretable algorithms for classification. It models the relationship between predictors and the probability of belonging to a certain class using the sigmoid function. Despite its simplicity, it performs remarkably well in many practical scenarios.

References

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- Analytics Vidhya
- Scikit-Learn Documentation
- Super Data Science